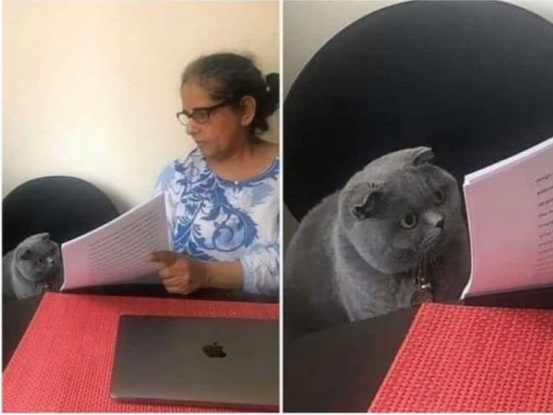


Session- 5

Deploy ON ECS

We'll start 9:05 PM

My manager showing me the code I wrote that took down the whole website



When you forget to **turn off** your AWS instance and receive your AWS bill



AGENDA

Brief Overview of AWS

To get our hands dirty → Deploy on EC2
→ Deploy on ECS

Other cloud computing platforms.

1. AWS - Amazon
2. GCP - Google
3. Azure - Microsoft
4. Digital Ocean
5. Heroku
6. IBM
7. Oracle

Aws provides → Raw Infra → EC2
Pre-Configured → ECS

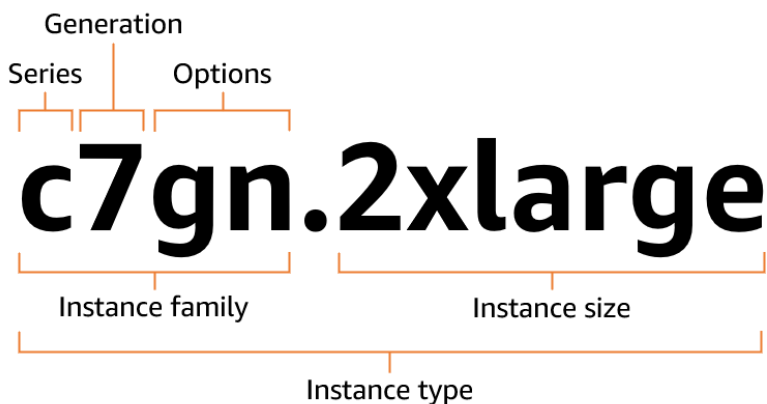
AWS Services

1. EC2 -> Elastic Compute Cloud (Blank/Un-configured cloud computers/servers).
2. S3 -> Simple Storage Service - Online Equivalent for Google Drive.
3. ECS -> Elastic Container Service.
4. ECR -> Elastic Container Registry (AWS equivalent to DockerHub).
5. SageMaker -> Google Colab inside Amazon AWS.
6. EMR -> Distributed Computing.
7. IAM -> Identity and Management - Control access throughout the AWS Infra.



EC2 instance types

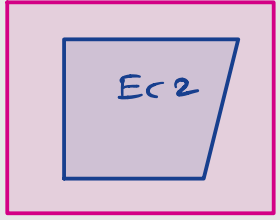
	General Purpose		Compute Optimized	Memory Optimized		Accelerated Computing	Storage Optimized		
Type	t2	m5	c5	r4	x1e	p3	h1	i3	d2
Description	Burstable, good for changing workloads	Balanced, good for consistent workloads	High ratio of compute to memory	Good for in-memory databases	Good for full in-memory applications	Good for graphics processing and other GPU uses	HDD backed, balance of compute and memory	SDD backed, balance of compute and memory	Highest disk ratio
Mnemonic	t is for tiny or turbo	m is for main or happy medium	c is for compute	r is for RAM	x is for xtreme	p is for pictures	h is for HDD	i is for IOPS	d is for dense



VIRTUAL PRIVATE CLOUD



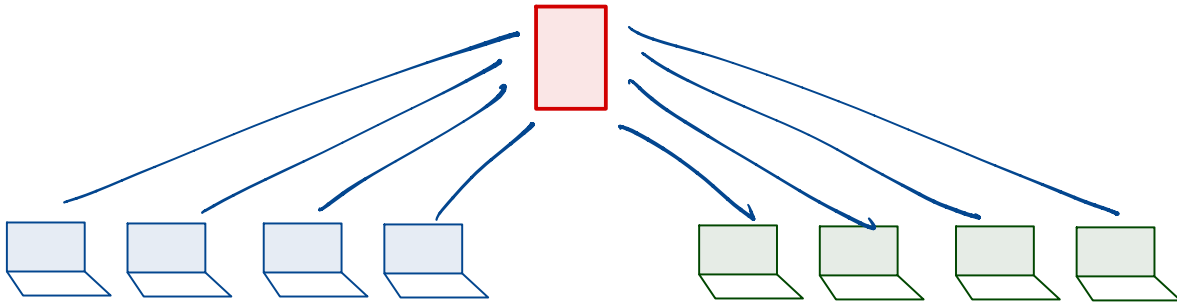
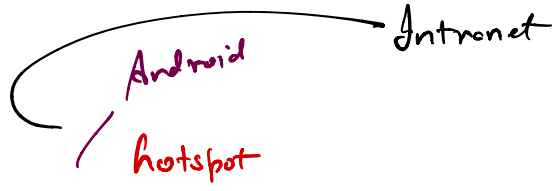
Security groups



Security groups



Internet gateway



EC2

- ① Rent a Server
For a long time

ECS

- ① Pay as you go
- ② Autoscaling

Spain → Ronda → Flat → 30K / month
Hotel → 5K / day

3 days → Hotel 15K

Autoscaling

- ① Horizontal



12 GB RAM
512 GB SSD



12 GB RAM
512 GB SSD



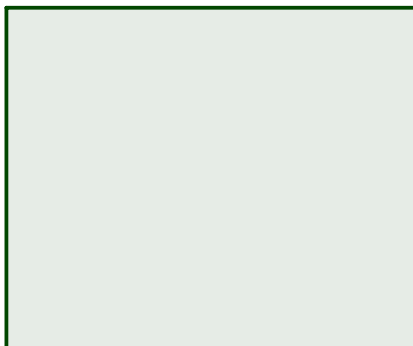
12 GB RAM
512 GB SSD

②

Vertical Scaling

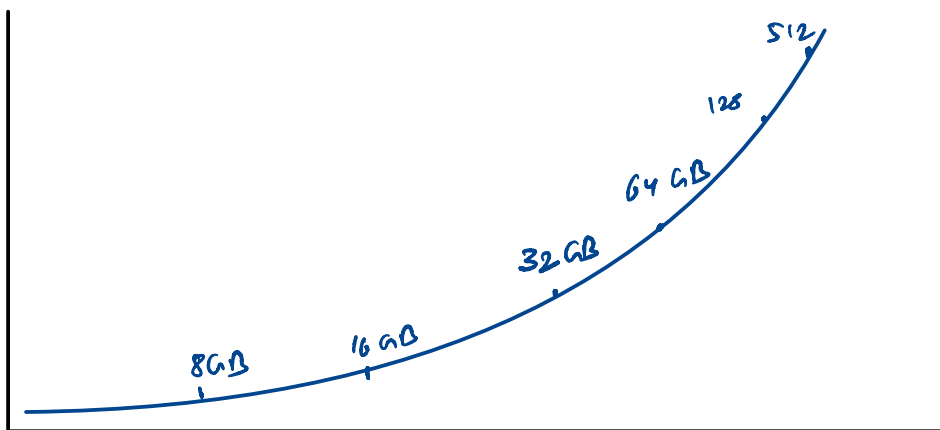


12 GB RAM
512 GB SSD



120 GB RAM
5120 GB SSD

Cost



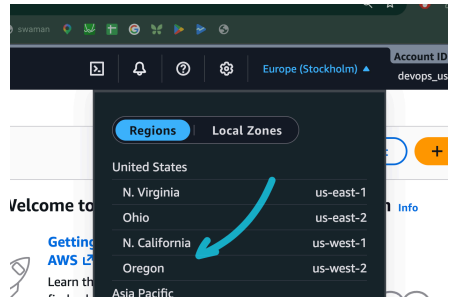
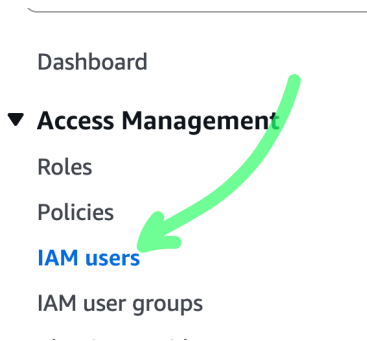
Configuration

Steps for Deployment

1. Container-ize the model.
2. Push container to registry.
3. Choose/setup online cloud server.
4. Pull image, deploy.
5. configure network access.

Same steps in context of Amazon ECS

1. Authenticate my computer with AWS using IAM.
2. Build Image and container-ize our model, and push to ECR.
3. Create Cluster, and task-definition.
4. Deploy.
5. Configure security groups.



```
sudo apt update
```

```
sudo apt install -y curl unzip
```

```
curl "https://awscli.amazonaws.com/awscli-exe-linux-x86_64.zip" -o "awscliv2.zip"
```

```
unzip awscliv2.zip
```

```
sudo ./aws/install
```

```
aws --version
```

Cluster →

Frontend

Backend

DB